

13	(a)	(i)	to align nuclei/protons to cause Larmor/precessional frequency to be in r.f. region	B1 B1	[2]
		(ii)	Larmor/precessional frequency depends on (applied magnetic) field strength knowing field strength enables (region of precessing) nuclei to be located by knowing the frequency	B1 M1 A1	[3]
	(b)		$E = 2.82 \times 10^{-26} \times B$ $6.63 \times 10^{-34} \times 42 \times 10^6 = 2.82 \times 10^{-26} \times B$	C1	
			$B = 0.99 \text{ T}$	A1	[2]